

Discrete Structures: Exercises

Chapter 1

1. (a) If $A = \{p, q, q, s\}$, $B = \{m, n, q, s, t\}$ and $C = \{m, n, p, q, s\}$. Then verify the associative property of the union below,

$$(m, n, q, s, t, p) = (p, q, s, m, n, t) \quad \text{(proven)}$$

$$A \cup (B \cup C) = (A \cup B) \cup C$$

- (b) By referring to the properties of set operations, show that:

$$A - B = A \cap B'$$

$$(X \cup Y) - Z = (X - Z) \cup (Y - Z)$$

$$(X \cup Y) \cap Z' = (X \cap Z') \cup (Y \cap Z') = (X - Z) \cup (Y - Z) \quad \text{(proved)}$$

- (c) Given $A = \{2, 3, 4, 5\}$ and $B = \{4, 16, 23\}$, $a \in A$, $b \in B$, find the set of ordered pairs such that $a^2 < b$

- (d) There are 35 students in the art class and 57 students in the science class. Find the number of students who are either in art class or in science class.

- i. When two classes meet at different hours and 12 students are enrolled in both activities.
ii. When two classes meet at the same hour.

$$(ii) \quad 35 + 57 = 92$$

2. (a) Consider the statement:

"If you try hard and you have a talent then you will get rich"

- i. Translate the statement into logic symbols. Use p , q , and r to represent the propositions. Clearly state which statement is p , q , and r .
ii. Suppose you found out that the statement was a lie even you try hard. What can you conclude?
iii. If you are rich but you do not try hard or have talent. Was the statement true or false? Support your conclusion.

- (b) Use truth table to check if the compound propositions A and B are logically equivalent.

$$A = \neg(p \vee (q \wedge (r \rightarrow p)))$$

$$B = \neg p \wedge (q \rightarrow r)$$

logically
equivalent

- (c) Let $P(x, y) : x + 2y = 1$, where the domain of discourse is the set of all integers. Determine the truth value of the following statements. Give the value of x and y that make the statement TRUE or FALSE.

- i. $\neg \forall x \forall y P(x, y)$
ii. $\exists x \exists y P(x, y)$

- (d) Proof that if x is an odd integer and y is an even integer then $x^2 - 2y$ is an odd integer using direct proofing.

$$x = 2k + 1 \quad y = 2m$$

$$(2k+1)^2 - 2(2m)$$

1

$$4k^2 + 1 + 4k - 4m$$

$$= 4k^2 + 1 = \text{odd (proven)} \\ + 4m$$

$$A + S - \text{Ans} \\ \text{Ans} = 35 + 57 - 12 \\ \text{Ans} = 12$$

$$A = \neg(p \vee (q \wedge (r \rightarrow p)))$$

$$B = \neg p \wedge (q \rightarrow r)$$

F T T

p	q	r	$\neg(p \vee (q \wedge (r \rightarrow p)))$	$\neg p \wedge (q \rightarrow r)$
T	T	T	F	F
T	T	F	F	F
T	F	T	F	F
T	F	F	F	F
F	T	T	F	T
F	T	F	F	T
F	F	T	T	F
F	F	F	T	F

$$2n_1 + 1 = 2n_2 + 1$$

$$n_1 = n_2$$

2. (a) Consider the statement:

“If you try hard and you have a talent then you will get rich “

- i. Translate the statement into logic symbols. Use p , q , and r to represent the propositions. Clearly state which statement is p , q , and r .
- ii. Suppose you found out that the statement was a lie even you try hard. What can you conclude?
- iii. If you are rich but you do not try hard or have talent. Was the statement true or false? Support your conclusion.

(i) $p = \text{try hard}$
 $q = \text{have talent}$
 $r = \text{get rich}$
 $(p \wedge q) \rightarrow r$

(ii) $(p \wedge q) \rightarrow r$
 $(T \wedge T) \rightarrow F$
conclusion

假的
 $\neg(p \wedge q) \rightarrow T$
 $\neg p \vee \neg q \rightarrow r$
 $F \vee F \rightarrow T$
 $F \rightarrow T$
statement is true

(iii) $\neg(p \vee q) \rightarrow r$

$\neg p \wedge \neg q \rightarrow r$
 $F \wedge F \rightarrow T$

statement
is true

(c) Let $P(x, y) : x + 2y = 1$, where the domain of discourse is the set of all integers. Determine the truth value of the following statements. Give the value of x and y that make the statement TRUE or FALSE.

i. $\neg \forall x \forall y P(x, y)$

ii. $\exists x \exists y P(x, y)$

$\neg P(x,y)$ (i) $x+2y \neq 1$ ^{not all $P(x,y) = x+2y=1$}

true

example (1,2)

$$1+2(2) \quad 5 \neq 1$$

(ii) $\exists x \exists y P(x,y)$ some $P(x,y) = x+2y=1$
example (-3,2)

$$-3+2(1)$$

$$P(x,y) \quad x+2y=1$$

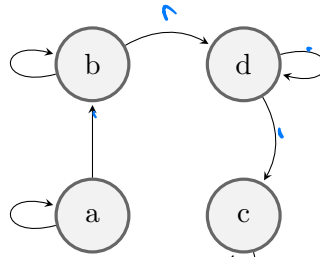
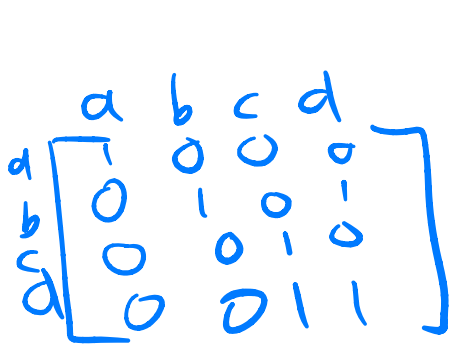
true

$$(-3)+2(2)=1$$

$$-3+4=1$$

Chapter 2

3. Given the digraph of relation R as in Figure



(a) List the elements of R as a set of ordered pairs.

(b) What is matrix representation of R ?

(c) Is the relation R antisymmetric, asymmetric or/and partial order relation? Give justification for each of your answers.

(d) Determine which of the relations are functions from $A = \{1, 2, 3\}$ to $B = \{4, 6, 8, 10\}$. Give justifications for each of your answers.

i. $R_1 = \{(1, 4), (2, 8), (3, 8)\}$

ii. $R_2 = \{(1, 4), (1, 8), (2, 6), (3, 10)\}$

iii. $R_3 = \{(1, 8), (3, 4)\}$

(e) Let $X = \{-3, 0, 3\}$ and $Y = \{-9, 0, 9\}$. For each $x \in X$, define the functions $v : X \rightarrow Y$ and $w : X \rightarrow Y$ by

$$v(x) = x^2, \quad w(x) = 3x$$

Determine if v and w are one-to-one, onto Y , and/or bijection.

(f) Let f and g be functions from the positive integers to the positive integers defined by the equations,

$$f(x) = 2x + 3, \quad g(x) = \frac{x}{5}$$

i. Find the inverse of $f(x)$ and $g(x)$.

ii. Find the compositions $f \circ g \circ f$.

Chapter 3

4. (a) Suppose there are five people in a corporate army dinner, including the captain and two vice-captain that are sitting around a circular table

i. How many ways for these people to be seated around the table.

ii. If the captain and both vice-captains should be seated next to each other.

(b) In a camping site, five people are staying in one camp with five allocated beds. The head of each camp must not sit next to their assistant to ensure safety of the camp in case of emergency. In that case, how many ways can you allocate the beds to all people in each camp?

$C(r)$
 $C(n+r-1, r)$
 $C(10+6-1, 6)$
 ${}^{15}C_6 = 5005$ ways

Dr. Tarmizi Adam

$$n=10$$

$$r=6$$

- (c) Suppose there are 10 types of chocolate that are available in a chocolate store. How many ways for you to buy half a dozen chocolate:
- If there are no restrictions.
 - If there at least 4 hazelnut flavoured chocolate.
 - If there are no two chocolates of the same type.
- (d) In a football match, there are thirteen players that show up to play for the next game.
- How many ways are there to choose 11 players to take start the game?
 - How many ways are there to assign 11 positions from the pool of 13 players?
 - Three players from 13 that showed up are woman. How many ways to choose 11 players to start the game if at least one player must be a woman?

$${}^{10}C_6 = 210 \text{ ways}$$

References

$(c)(ii)$ $C(10+2-1, 2) = {}^{11}C_2 = 55 \text{ ways}$

$$n=10$$

$$r=2$$

$(c)(iii)$ ${}^{10}C_6$

- (d) In a football match, there are thirteen players that show up to play for the next game.
- i. How many ways are there to choose 11 players to take start the game?
 - ii. How many ways are there to assign 11 positions from the pool of 13 players?
 - iii. Three players from 13 that showed up are woman. How many ways to choose 11 players to start the game if at least one player must be a woman?

$$d(i) \quad {}^{13}C_{11} = 78 \text{ ways}$$

$$C(i) \quad {}^{13}P_{10} = 13 \times 12 \times 11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 = 3113510400 \text{ ways}$$

$$(ii) \quad \binom{3}{9} x^{10} C_{10} + \binom{3}{2} x^{10} C_9 + \binom{3}{3} x^{10} C_8$$